U.S. Geological Survey Benchmark Glacier Project

Introduction

Climate change alters the fate and state of Earth's glaciers. Mountain glaciers, particularly in Alaska, contribute significantly to sea level rise and its global implications (Zemp and others, 2019). Before reaching the sea, glacier melt provides freshwater for human consumption, agriculture, recreation, and hydropower. Glacier change also affects mountain hazards (Harrison and others, 2018).

Glaciers release melt water during the late summer season, which sustains streamflow between spring snow melt and fall rain (Curran and Biles, 2021). Glaciers are biologically linked to their downstream ecosystems, providing cold water critical for salmon (Wainwright and Weitkamp, 2013) and aquatic habitats (Giersch and others, 2017). Alterations to seasonal stream flow, water chemistry, and temperature (Déry and others, 2009; Bergstrom and others, 2021) caused by glacier loss pose challenges for water resource managers, ecosystems, and downstream communities who depend on these reservoirs of frozen water.

For decades, key glaciers situated across western North America have been closely monitored (fig. 1). These consistent field measurements have provided valuable "benchmarks" (Fountain and others, 1997) to understand the physical processes that link glaciers to climate, ecosystems, and hydrology (Mathes, 1939).

Figure 1. Photograph showing the Wolverine Glacier research hut, Alaska. Photograph by Louis Sass, U.S. Geological Survey.

Decades of Data

Five U.S. Geological Survey (USGS) Benchmark Glacier sites span mid to high latitudes and represent climatically distinct regions, from inland to coastal, providing a broad sample of climate response across the continent (fig. 2). Four of the glaciers are considered reference glaciers in the World Glacier Monitoring Service's internationally coordinated glacier monitoring network (Zemp and others, 2015), with data used regularly in international climate assessment reports (Pfeffer and others, 2014). The long-term record of glacier change captures seasonal and year-to-year variability.

In 2019, the USGS Benchmark Glacier Project consolidated field practices and data processing methods (O'Neel and others, 2019). This ensures standardized, intercomparable estimates of glacier change across the continent, which boosts scientific value beyond individual glacier monitoring efforts.

Overall, insight gained from decades of USGS glaciology research facilitates glacier change projections, which guide sea level and water resource management strategies.

Modern Measurements

USGS scientists visit the same Benchmark Glacier sites (fig. 2) every spring and fall to make direct measurements of snow accumulation and snow and ice melt. USGS scientists collect snow accumulation data in winter using snow pits and snow cores (fig. 3), while summer melt is measured via annual visits to stakes installed in the glacier surface. Images captured from space allow scientists to document glacier change over entire mountain ranges. Collectively, these data allow scientists to closely track annual changes in both climate and glacier mass. Direct measurements are costly and logistically challenging, but necessary for continuity and validation of newer remote sensing methods of tracking glacier change.

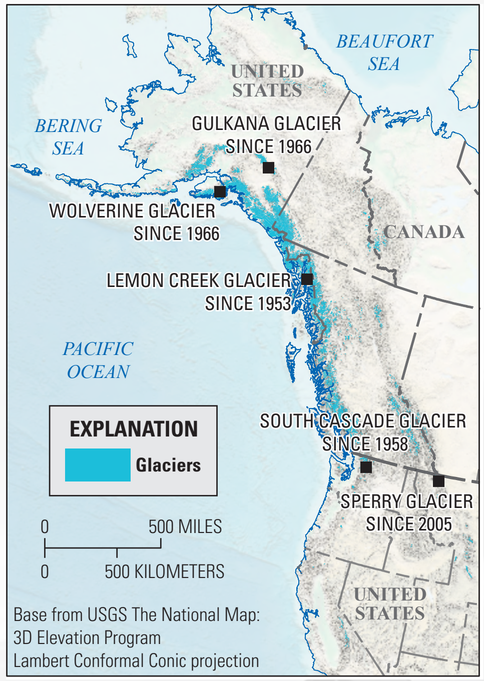

Figure 2. Map showing the USGS Benchmark Glaciers (black squares) that span glacierized regions (blue) of North America (glacier extent data from the RGI Consortium, 2015).

Conclusion

USGS scientists are expanding the relevance of the Benchmark Glacier network as they layer complementary new technologies atop decades of, and on-going direct glaciological studies. Leveraging high performance computing and satellite imagery, scientists monitor glaciers across diverse climatic zones, providing the holistic understanding of glacier change needed for adaptive water resource strategies.

Understanding glacier change on regional scales using direct field measurements and remotely sensed data allows the USGS Benchmark Project to provide actionable information for downstream science and stakeholders. This includes other scientists focused on species habitat, ecosystem function, glacier geophysics, oceanography, and resource management.

Figure 3. Scientists collect snow accumulation and density data at Gulkana Glacier on April 30, 2020. Photograph by the U.S. Geological Survey.

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For More Information

Contact USGS Northern Rocky Mountain Science Center, P.O. Box 169, West Glacier, MT 59936. Or visit the Northern Rocky Mountain Science Center website at https://www.usgs.gov/centers/norock.

By Caitlyn Florentine and Lisa A. McKeon

Banner The Wolverine Glacier, Alaska in September 2019. Photograph by Caitlyn Florentine, U.S. Geological Survey.

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